

Exhibit 4: EAA Equity Index Daily Returns Grouped by Size of Return

Bin	Cumulative Percentage of Sample Trading Days (%)	Daily Return (%) between		Number of Observations
		Lower Bound	Upper Bound	
1	5	−4.108	−1.416	63
2	10	−1.416	−0.876	63
3	15	−0.876	−0.629	63
4	20	−0.629	−0.432	63
5	25	−0.432	−0.293	63
6	30	−0.293	−0.193	63
7	35	−0.193	−0.124	62
8	40	−0.124	−0.070	63
9	45	−0.070	−0.007	63
10	50	−0.007	0.044	63
11	55	0.044	0.108	63
12	60	0.108	0.173	63
13	65	0.173	0.247	63
14	70	0.247	0.343	62
15	75	0.343	0.460	63
16	80	0.460	0.575	63
17	85	0.575	0.738	63
18	90	0.738	0.991	63
19	95	0.991	1.304	63
20	100	1.304	5.001	63

Because of the continuous nature of returns, it is not likely for a return to fall on the boundary for any bin other than the minimum (Bin = 1) and maximum (Bin = 20).

Using the data in Exhibit 4, complete the following tasks:

1. Identify the 10th and 90th percentiles.

Solution:

The 10th and 90th percentiles correspond to the bins or ranked returns that include 10 percent and 90 percent of the daily returns, respectively. The 10th percentile corresponds to the return of −0.876 percent (and includes returns of that much and lower), and the 90th percentile corresponds to the return of 0.991 percent (and lower).

2. Identify the first, second, and third quintiles.

Solution:

The first quintile corresponds to the lowest 20 percent of the ranked data, or −0.432 percent (and lower).

The second quintile corresponds to the lowest 40 percent of the ranked data, or −0.070 percent (and lower).

The third quintile corresponds to the lowest 60 percent of the ranked data, or 0.173 percent (and lower).